

Utku Okur

Contact information

Faculty of Mathematics and Computer Science, TU Bergakademie Freiberg
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Born: January 5, 1992 — Izmir, Turkey

Current position

2026– Postdoc, Institute of Discrete Mathematics and Algebra, TU Bergakademie Freiberg, Germany

Areas of interest

Discrete mathematics, algebraic graph theory, combinatorics, formal verification (Lean 4)

Education

- 2020–2025 PhD in Mathematics, University of South Carolina, USA. Advisor: Joshua Cooper. Thesis: *Coefficients of the characteristic polynomial of a hypergraph and their combinatorial properties*
- 2018–2020 MSc in Mathematics, Bilkent University, Turkey. Advisor: Laurence Barker. Thesis: *Mackey decomposition for Brauer pairs*
- 2013–2018 BSc in Mathematics and BA in Philosophy, Boğaziçi University, Turkey

Research publications

8. Aida Abiad, Joshua Cooper, Utku Okur, *A general switching method for constructing E -cospectral hypergraphs*, Preprint (2026). [arXiv:2604.06016](https://arxiv.org/abs/2604.06016)
7. Ann Clifton, Éva Czabarka, Audace A. V. Dossou-Olory, Kevin Liu, Sarah Loeb, Utku Okur, László Székely, Kristina Wicke, *Reconstruction of Planar Tanglegrams*, In preparation (25+).
6. Ann Clifton, Éva Czabarka, Kevin Liu, Sarah Loeb, Utku Okur, László Székely, Kristina Wicke, *Reconstruction of caterpillar tanglegrams*, Discrete Applied Mathematics 378 (2026), 647–661. [arXiv:2501.13670](https://arxiv.org/abs/2501.13670)
[doi:10.1016/j.dam.2025.09.022](https://doi.org/10.1016/j.dam.2025.09.022)
5. J. Cooper, U. Okur, *Multigraphs with Unique Partition into Cycles*, Submitted (25+) to Discussiones Mathematicae Graph Theory. [arXiv:2504.08083](https://arxiv.org/abs/2504.08083)
[doi:10.7151/dmgt.2627](https://doi.org/10.7151/dmgt.2627)
4. J. Cooper, U. Okur, *Partition Lattice of an Eulerian Digraph*, Submitted (25+) to Discrete Mathematics. [arXiv:2502.00867](https://arxiv.org/abs/2502.00867)

3. Ann Clifton, Éva Czabarka, Audace A. V. Dossou-Olory, Kevin Liu, Sarah Loeb, Utku Okur, László Székely, Kristina Wicke, *Decks of rooted binary trees*, European Journal of Combinatorics 124 (2025), 104076. [arXiv:2311.02255](#)
[doi:10.1016/j.ejc.2024.104076](#)
2. J. Cooper, K. Guo, U. Okur, *Characteristic Polynomials and Hypergraph Generating Functions via Heaps of Pieces*, Submitted (26+) to Linear and Multilinear Algebra. [arXiv:2411.03567](#)
1. J. Cooper, U. Okur, *Polynomial reconstruction problem for hypergraphs*, Linear Algebra and its Applications 692 (2024), 37–61. [arXiv:2312.16152](#)
[doi:10.1016/j.laa.2024.03.029](#)

GitHub repositories

- Hypergraph calculations:
<https://github.com/utkuokur/Hypergraph-Programs>
- Tanglegram calculations:
https://github.com/utkuokur/binary_trees_and_tanglegrams

Other writings

- Vaibhav Bajpai, Nathan Bowler, Johannes Carmesin, Utku Okur, *10 Scalable Challenges in Combinatorics: A Public Benchmark Formalised in Lean*, Extended abstract, AI4Math workshop at CICM 2026, Ljubljana; informal proceedings, [PDF](#).
- Vaibhav Bajpai, Johannes Carmesin, Utku Okur, *A Lean 4 Formalisation of OpenAI's Proof of the Cycle Double Cover Conjecture*, Lean code: [GitHub](#), [doi:10.5281/zenodo.21410822](#). See also J. Carmesin, *The cycle double cover theorem*, The Matroid Union, July 2026.
- Ann Clifton, Éva Czabarka, Kevin Liu, Sarah Loeb, Utku Okur, László Székely, Kristina Wicke, *Universal rooted phylogenetic tree shapes and universal tanglegrams*, arXiv preprint (2023), [arXiv:2308.06580](#); not intended for publication, since a better result exists; see Gawrychowski, Kuhn, Łopuszański, Panagiotou, Su, SODA 2018, [doi:10.1137/1.9781611975031.166](#).

Talks

2024	Discrete Math and Combinatorics Seminar, University of South Carolina, Columbia/SC.
2023	<i>Invited Speaker</i> : Joint Mathematics Meetings, AMS Special Session on Trees in Many Contexts, Boston/MA.
2023	Graduate Mini-Conference in Discrete Mathematics, University of South Carolina, Columbia/SC.
2023	Discrete Math and Combinatorics Seminar, University of South Carolina, Columbia/SC.

Workshops attended

- 2026 [Workshop on Artificial Intelligence for Mathematics \(AI4Math\)](#), Conference on Intelligent Computer Mathematics (CICM 2026), September 21, 2026, Ljubljana.
- 2022 Trees in Many Contexts, Mathematics Research Community (MRC) Conference, June 5–11, 2022, Boston.

Grants, scholarships & awards

- 2013–2018 TUBITAK Scholarship for Fundamental Sciences
- 2019 Orhan Alisbah Fellowship Award

Teaching summary

- 2018–2020 Teaching Assistant at Bilkent University, Turkey
- 2019–2020 Research Assistant at Middle Eastern Technical University (METU), Turkey
- 2020–2021 Teaching Assistant at the University of South Carolina (UofSC), USA
- 2021–2025 Instructor of Record at the University of South Carolina (UofSC), USA

Teaching experience

- 2020 *Teaching Assistant (TA)*: Math 142 – Calculus II
- 2021 *Instructor*: Math 115 (Precalculus Mathematics)
- 2022 *Instructor*: Math 115, Math 122 (Calculus for Business Administration & Social Sciences), Math 111 (Basic College Mathematics)
- 2023 *Instructor*: Math 115, Math 122, Math 241 (Vector Calculus)
- 2024 *Instructor*: Math 122, Math 111, Math 170 (Finite Mathematics)

Service

- 2024 *Organizer*: Discrete Math and Combinatorics Seminar, University of South Carolina
- 2023–2025 *Referee*: Linear and Multilinear Algebra, The Electronic Journal of Linear Algebra, Theoretical Computer Science, Discrete Mathematics